



CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE PROGRAMME (CU-FYUGP)

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	PHYSICS OF THE HUMAN BODY				
Type of Course	Major Elective (SPECIALIZATION III: PHYSICS IN BIOLOGY)				
Semester	V				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Newtonian mechanics.				
Course Summary	This course analyses the human body from the viewpoint of mechanics and its static and dynamic equilibrium. The effects of collisions on human body, leading to fractures are explored. The significance of muscles of the human body is also analysed.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand & apply the laws of mechanics to the human body w.r.to its static equilibrium.	U, Ap	F	Instructor-created exams / Quiz
CO2	Understand dynamic equilibrium of human body.	U	F	Instructor-created exams / Quiz
CO3	Understand and analyse the effects of collision of human body from a mechanical force viewpoint.	U, An	F	Instructor-created exams / Quiz
CO4	Gain basic knowledge about various supporting structures of bones, a.k.a	U	F	Instructor-created exams / Quiz

	Ligaments, Tendons, Cartilage and how energy is stored in them.			
CO5	Basic understanding of fractures from mechanical force viewpoint.	U, An	F	Instructor-created exams / Quiz
CO6	Gain ideas about muscle and muscle activity from a mechanical viewpoint.	U, Ap	F	Instructor-created exams / Quiz
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	Static Equilibrium of the Body		12	15
	1	Review of Forces, Torques, and Equilibrium. (Section 2.1)	1	
	2	Statics: Motion in One Plane and Levers. (Section 2.2)	2	
	3	Statics in the Body. The lower arm and hip as examples. (Sections 2.3, 2.3.1 (Case 1, Case 2 only), 2.3.2 (variation of problem with cane to provide support on the left side not required).	2	
	4	Total body equilibrium. (Section 2.3.2)	2	
	5	Equilibrium of the individual body components. (Section 2.3.2)	2	
	6	Standing: Stability: overall stability, local stability. (Section 3.2)	2	
	7	Forces on the feet. (Section 3.2)	1	
Sections from References: Chapter 2 and 3 of Book 1				
II	Physical Aspects of Walking		8	15
	8	Kinematics of walking, Friction (Electromyographic activity of the muscles not required). (Section 3.3, 3.3.1, 3.3.3)	3	
	9	Energetics. Collisions of the human body: Kinematics of a collision, partially elastic collisions. (Section 3.3.4, 3.10, 3.10.1)	3	
	10	Consequences of collisions (upto & including calculation of GSI, p.178. Modification of GSI not required). (Section 3.10.2)	2	
Sections from References: Chapter 3 of Book 1				

III	Material Components of the Body		18	25
	11	Introduction to Bone. (Section 4.1, 4.1.1)	3	
	12	Ligaments and Tendons, Cartilage. (Section 4.1.2, 4.1.3)	1	
	13	Elastic Properties: Basic Stress–Strain Relationships. (Section 4.2.1)	3	
	14	Other Stress–Strain Relations, Bone Shortening. (Section 4.2.2, 4.2.3)	3	
	15	Energy Storage in Elastic Media. (Section 4.2.4)	3	
	16	Energy Storage in Tendons and Long Bones. (Section 4.2.4)	3	
	17	Bone Fractures: Modes of Sudden Breaking of Bones (Section 4.7, 4.7.1 up to and excluding Breaking of Bones by Bending).	2	
Sections from References: Chapter 4 of Book 1				
IV	Physical Aspects of Muscles		10	15
		Muscles, Skeletal Muscles in the Body. (Section 5, 5.1)	2	
	19	Types of Muscle Activity - The Structure of Muscles (upto electron micrograph figure showing banded myofilament structure (p.339)). (Section 5.1.1, 5.2)	2	
	20	Activating Muscles: Macroscopic View. (Section 5.3 only. Section 5.3.1 not required).	2	
	21	Muscle Strength and Evolution: Increasing Strength with Training.	2	
	22	Muscle Evolution with Age, Muscle Fatigue. (Section 5.11).	2	
Sections from References: Chapter 5 of Book 1				
V	Open Ended Module		12	
	Advanced features of walking, running, jumping, Avoiding Fractures and Other Injuries: Materials for Helmets			
	Sections Chapters 3 & 4: 3.3, 3.4, 3.5, 4.9 of Book 1			
Books and References:				
1. <i>Physics of the Human Body</i> , 2 nd Edition, Irving P. Herman, Springer International Publishing, 2016 (Book 1)				

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PS O5	PS O6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	2	1	0	1	0	0	2	2	0	0	1	0	1
CO 2	2	0	0	0	0	0	2	2	0	0	0	0	0
CO 3	2	0	0	0	0	0	2	2	0	0	0	0	1
CO 4	2	0	0	0	0	0	2	2	0	0	0	0	1
CO 5	2	0	0	0	0	0	2	2	0	0	0	0	0
CO 6	2	1	0	0	0	0	2	2	0	0	0	0	1

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/ Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	